

Privacy-Preserving Machine Learning on Web Browsing for Public Opinion

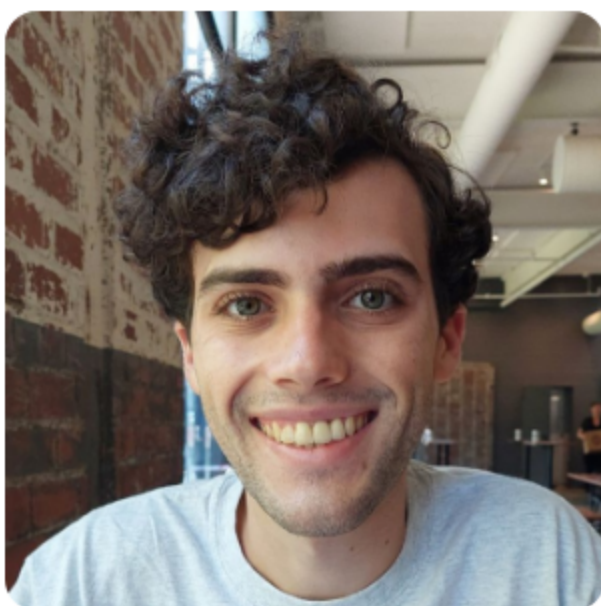
Sam Buxbaum

Lucas M. Tassis, Lucas Boschelli, Giovanni Comarela, Mayank Varia, Mark Crovella, Dino P. Christenson



CSCML 25

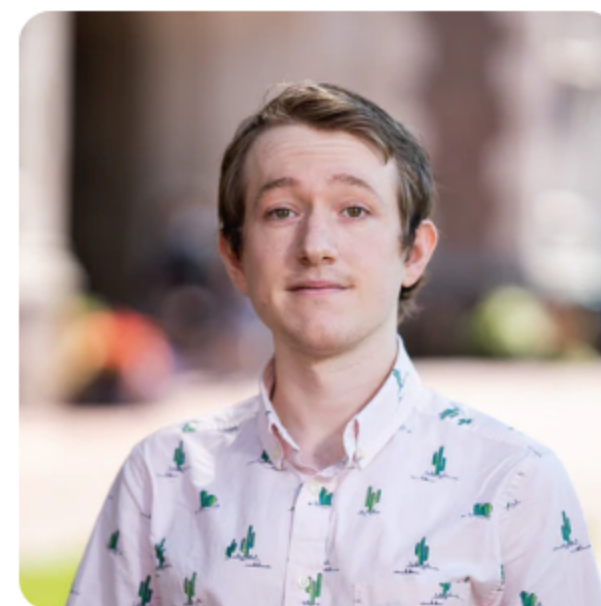
December 4, 2025



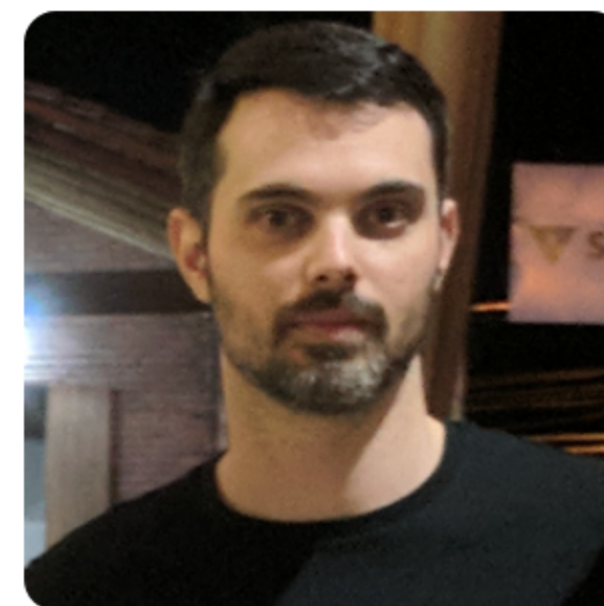
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Traditional Political Polling

- Data collection takes time
- Data collection is human-intensive
- Limited geographic and temporal coverage

West Virginia 2024 Presidential Election Polls



Harris vs. Trump

Source	Date	Sample	Harris	Trump	Other
Research America	8/30/2024	400 LV $\pm$ 4.9%	34%	61%	5%

Michigan 2024 Presidential Election Polls



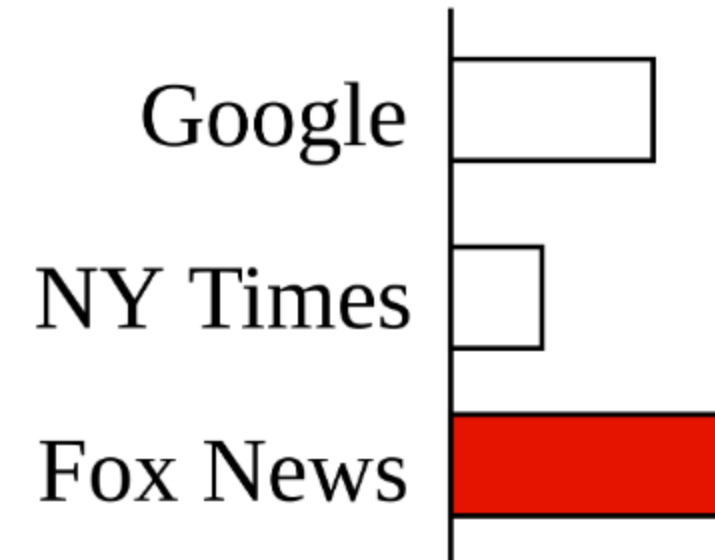
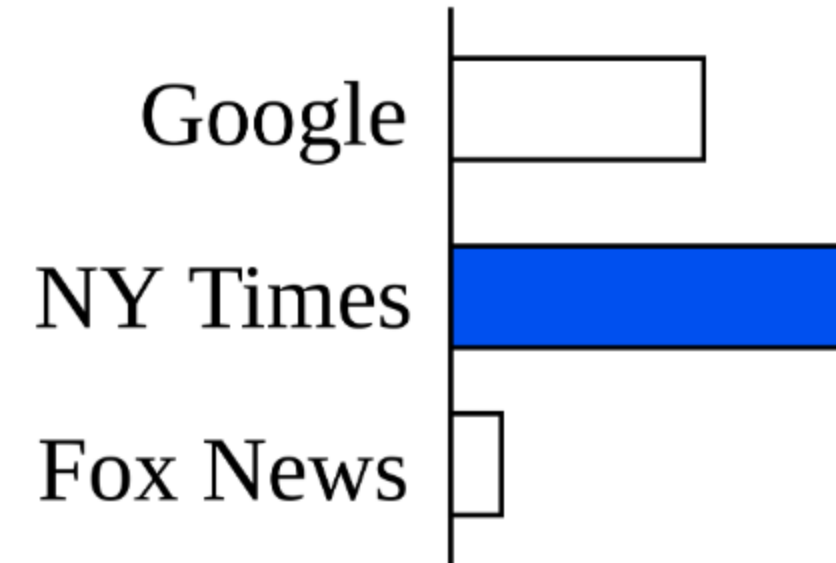
☐ Instantly compare a poll to prior one by same pollster

Harris vs. Trump

Source	Date	Sample	Harris	Trump	Other
Average of 23 Polls†			48.6%	46.8%	-
FAU / Mainstreet	11/04/2024	713 LV	49%	47%	4%
Emerson College	11/04/2024	790 LV $\pm$ 3.4%	50%	48%	2%
Research Co.	11/04/2024	450 LV $\pm$ 4.6%	49%	47%	4%
InsiderAdvantage	11/03/2024	800 LV $\pm$ 3.7%	47%	47%	6%
Trafalgar Group	11/03/2024	1,079 LV $\pm$ 2.9%	47%	48%	5%
MIRS / Mich. News Source	11/03/2024	585 LV $\pm$ 4%	50%	48%	2%
NY Times / Siena College	11/03/2024	998 LV $\pm$ 3.7%	47%	47%	6%
Morning Consult	11/03/2024	1,108 LV $\pm$ 3%	49%	48%	3%
AtlasIntel	11/02/2024	1,198 LV $\pm$ 3%	48%	50%	2%
Redfield & Wilton	11/01/2024	1,731 LV $\pm$ 2.2%	47%	47%	6%
The Times (UK) / YouGov	11/01/2024	942 LV $\pm$ 3.9%	48%	45%	7%
EPIC-MRA	11/01/2024	600 LV $\pm$ 4%	48%	45%	7%
Marist Poll	11/01/2024	1,214 LV $\pm$ 3.5%	51%	48%	1%
AtlasIntel	10/31/2024	1,136 LV $\pm$ 3%	49%	49%	2%
Echelon Insights	10/31/2024	600 LV $\pm$ 4.4%	48%	48%	4%
MIRS / Mich. News Source	10/31/2024	1,117 LV $\pm$ 2.5%	47%	49%	4%
UMass Lowell	10/31/2024	600 LV $\pm$ 4.5%	49%	45%	6%
Washington Post	10/31/2024	1,003 LV $\pm$ 3.7%	47%	46%	7%
Fox News	10/30/2024	988 LV $\pm$ 3%	49%	49%	2%
CNN	10/30/2024	726 LV $\pm$ 4.7%	48%	43%	9%
Suffolk University	10/30/2024	500 LV $\pm$ 4.4%	47%	47%	6%

Web Browsing for Political Polling

- Can website visits predict political leanings?
- Example - news websites
- More data
- Fully automated



Prior Work

- Web browsing behavior can predict voting results
- Quantifying the 'Comey letter' (Comarela et al.)
- Social media referrals are the best signal
- Used large plaintext dataset

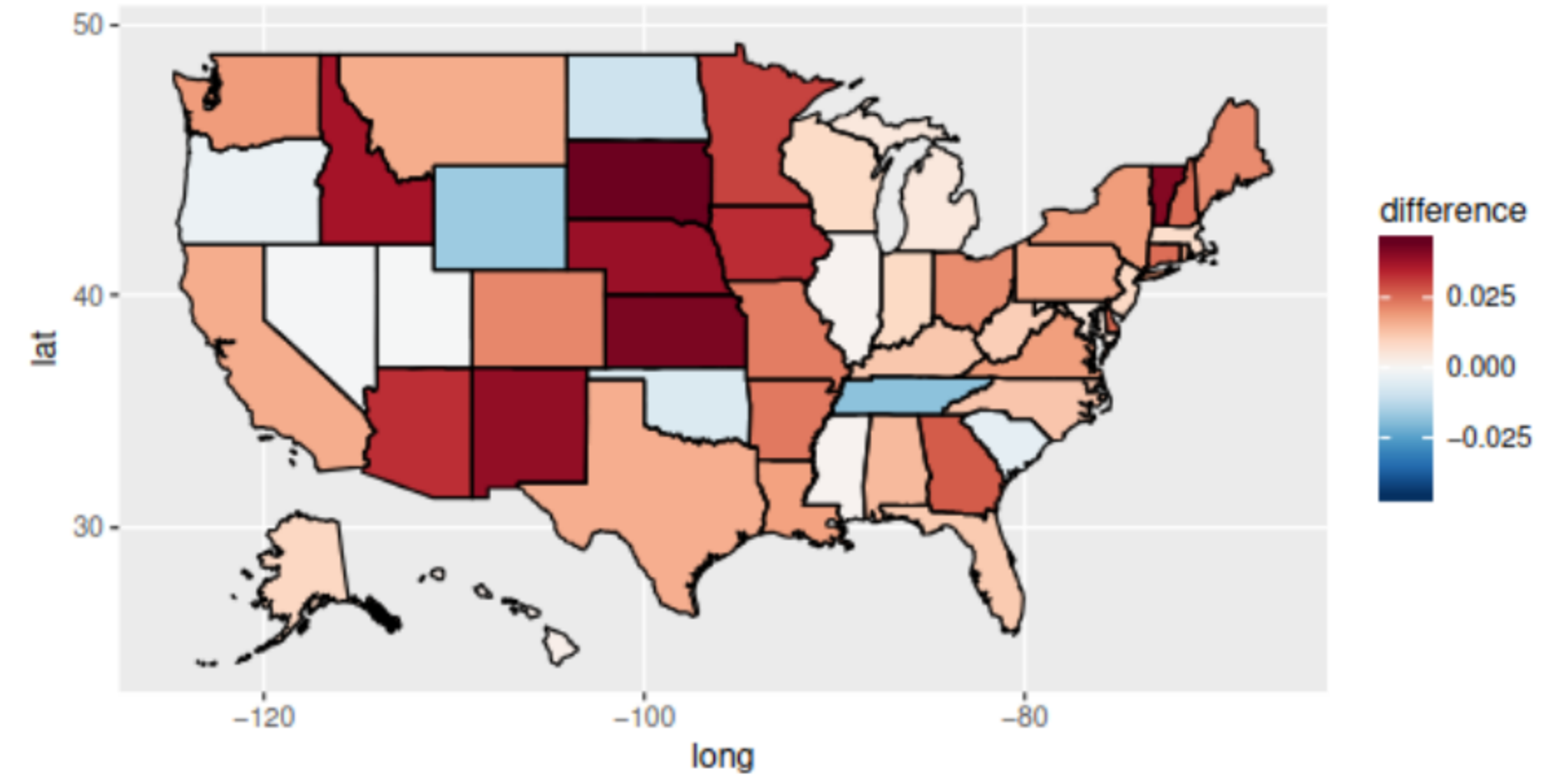
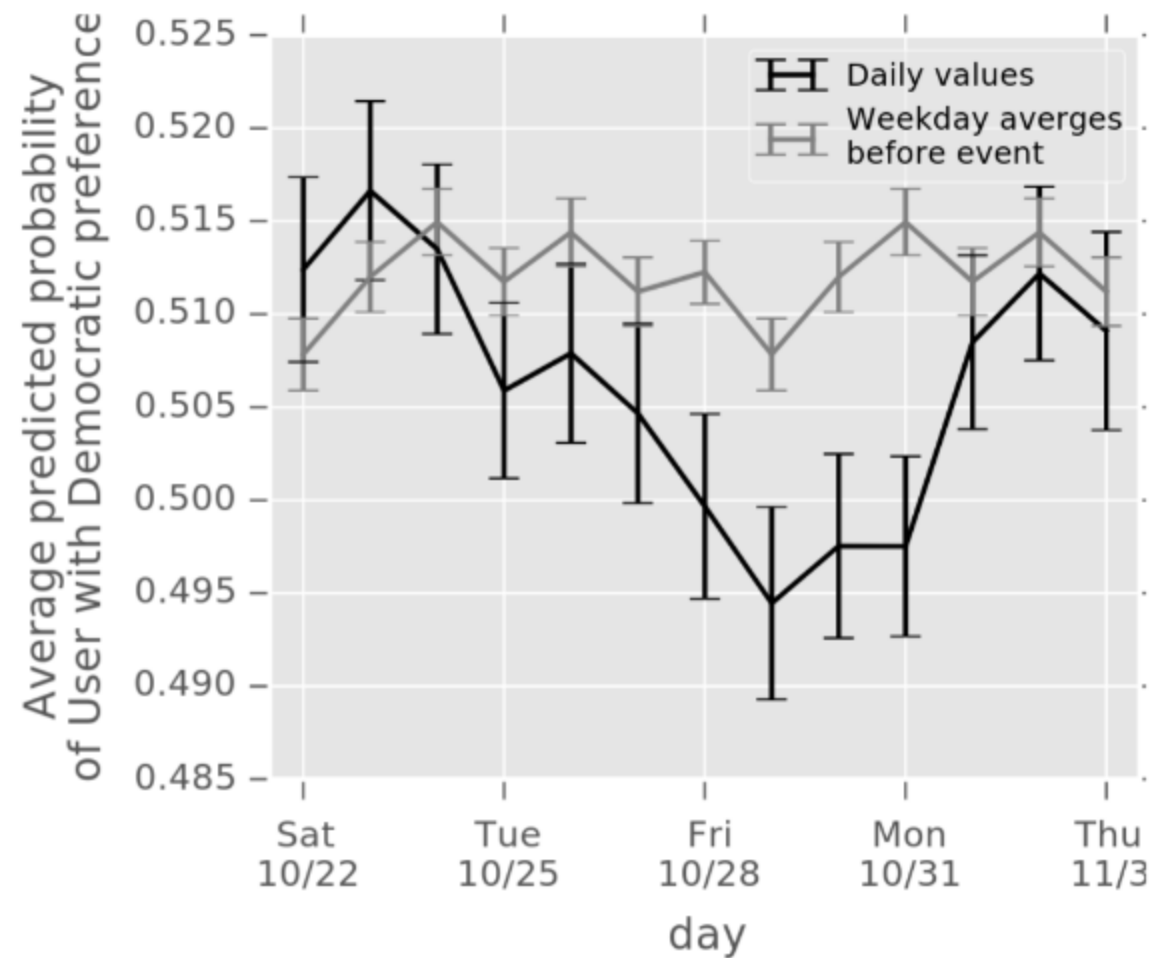
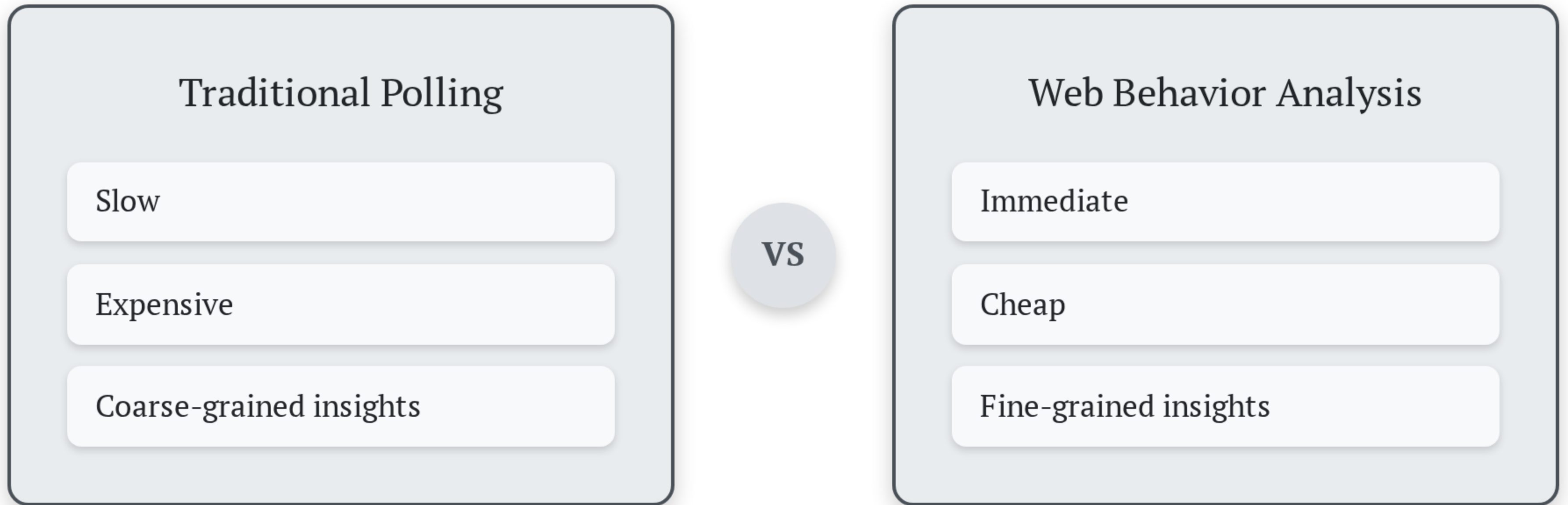


Figure 8: Impact of the 'Comey letter' at the state level.

Two Approaches to Political Polling



What about privacy?

Our Contributions

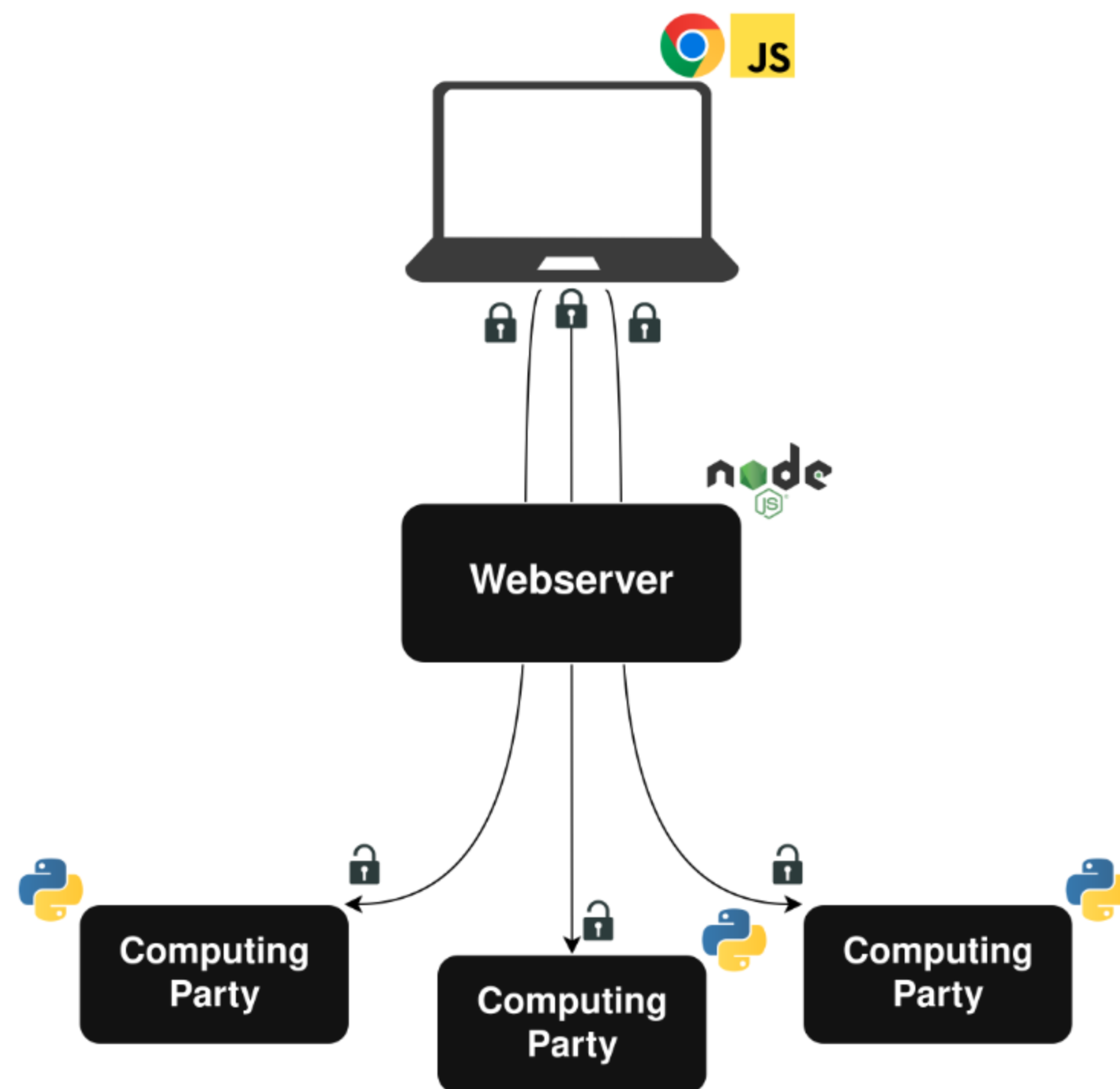
- We built a system for securely predicting political preferences from web browsing data
- We deployed the system during the 2024 U.S. presidential election
- We collected and analyzed data from almost 8000 unique users
- All analysis took place under MPC

System Design

Client Plugin

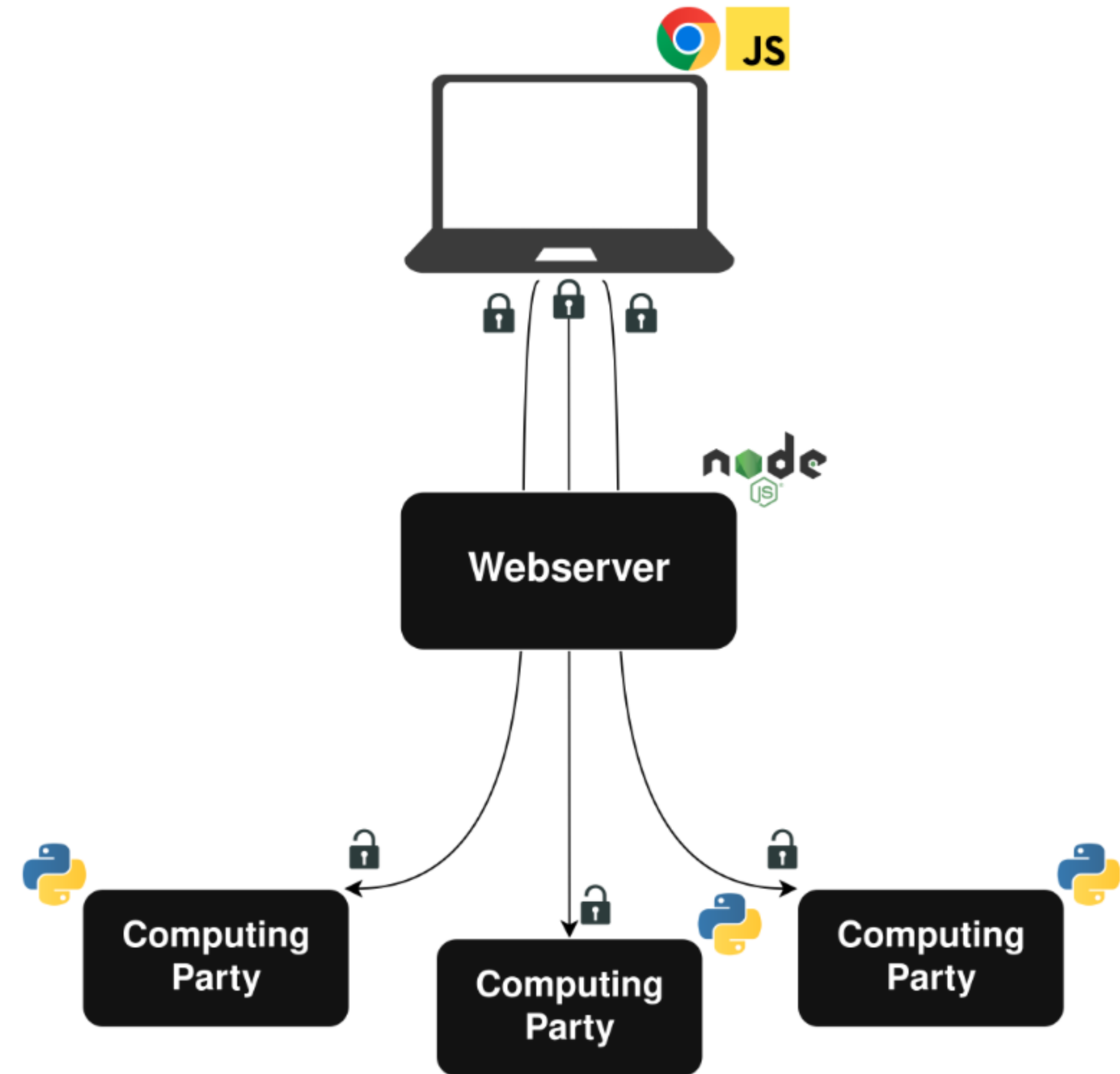
Webserver

MPC Backend



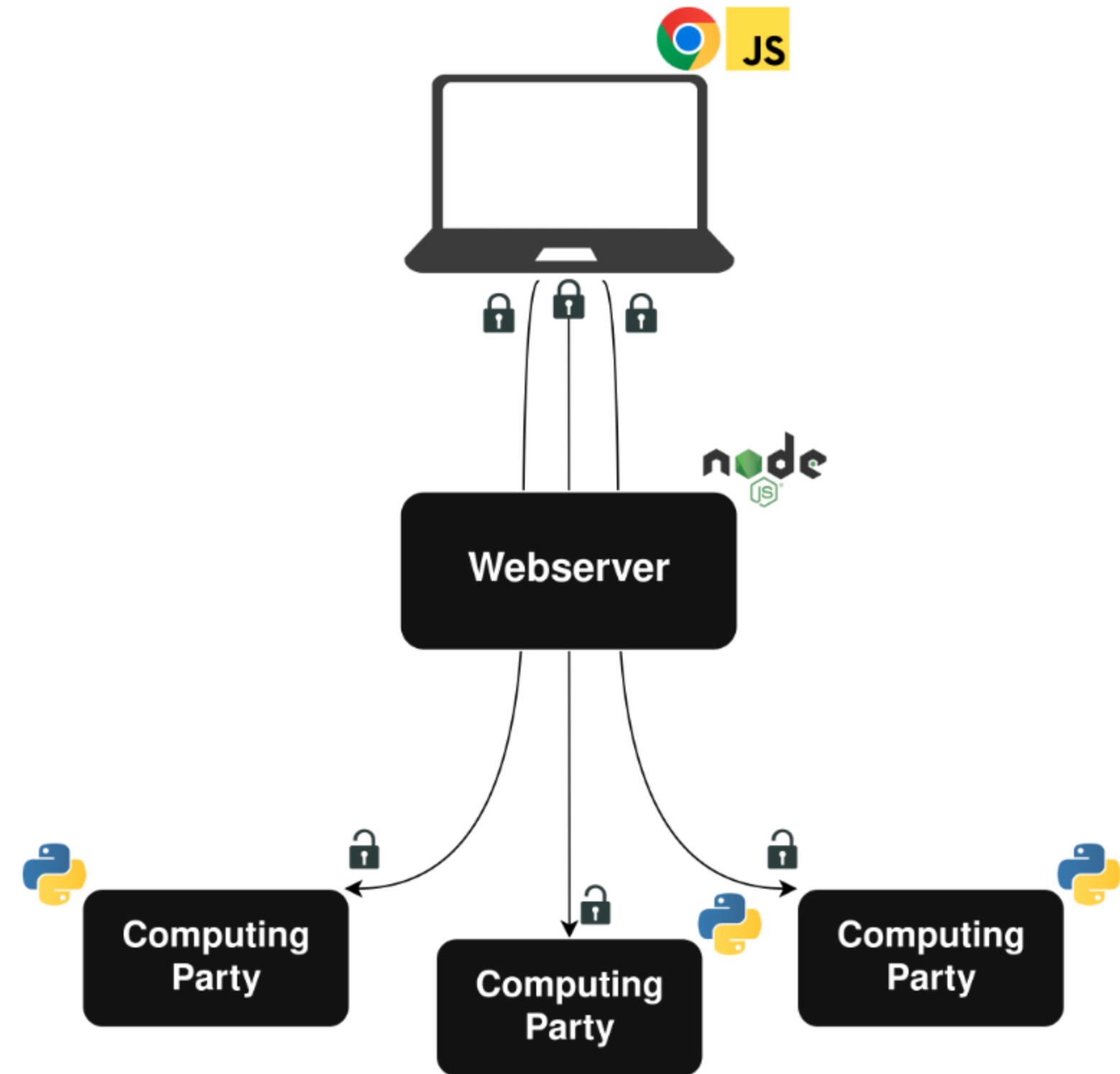
Client Plugin

- Custom-built Chrome plugin to monitor browsing
- Daily data uploads of secret-shared histograms
- Client-side secret sharing and encryption
- Implementation is open source



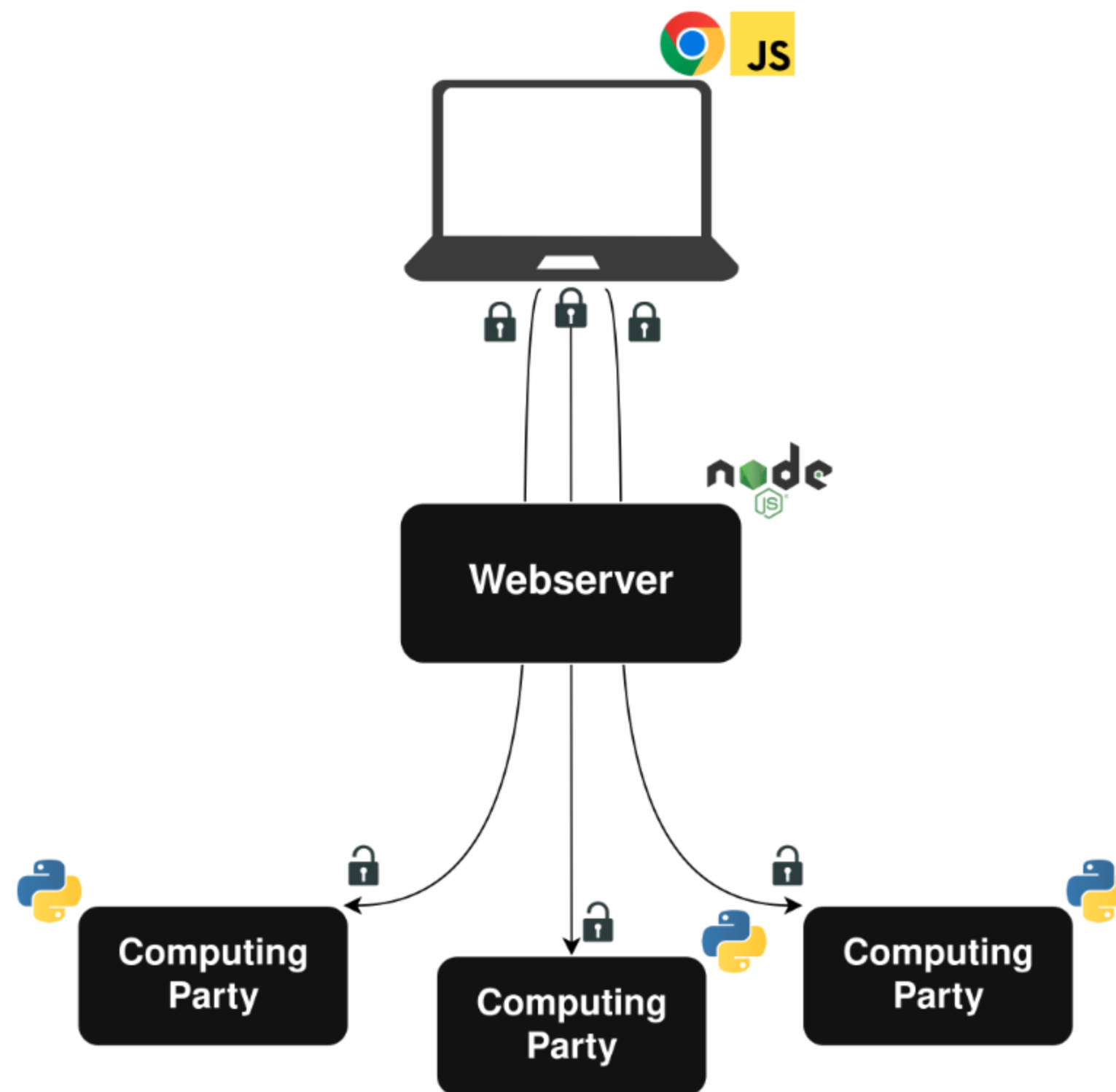
Webserver

- Simplifies interaction with clients
- Collects basic metadata
- Never sees any private data



MPC Backend

- Three party computation with an honest majority
- We used and augmented the CrypTen library
- We implemented an algorithm for LLP under MPC



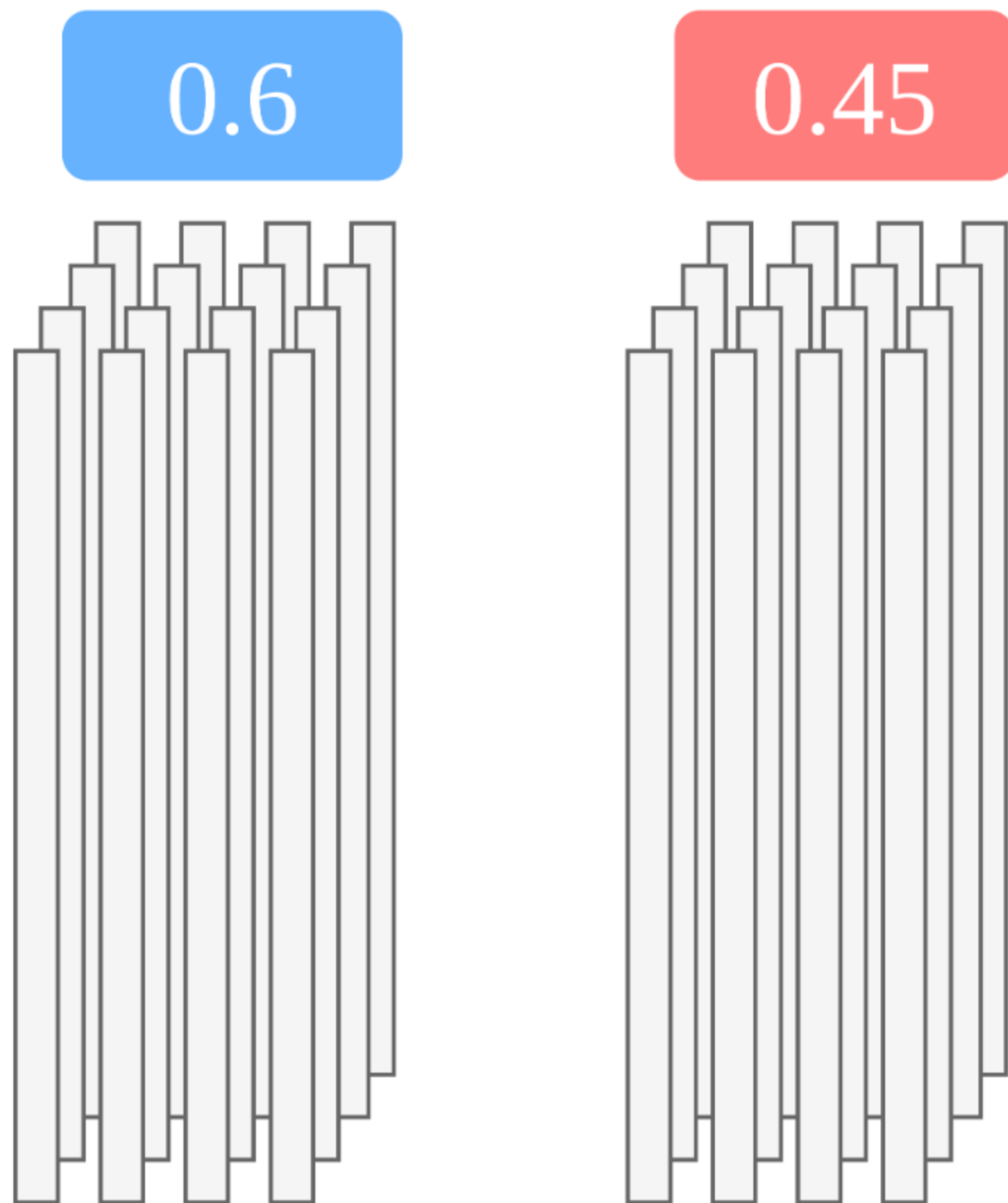
The Learning Process

We're working with the learning from label proportions (LLP) problem.

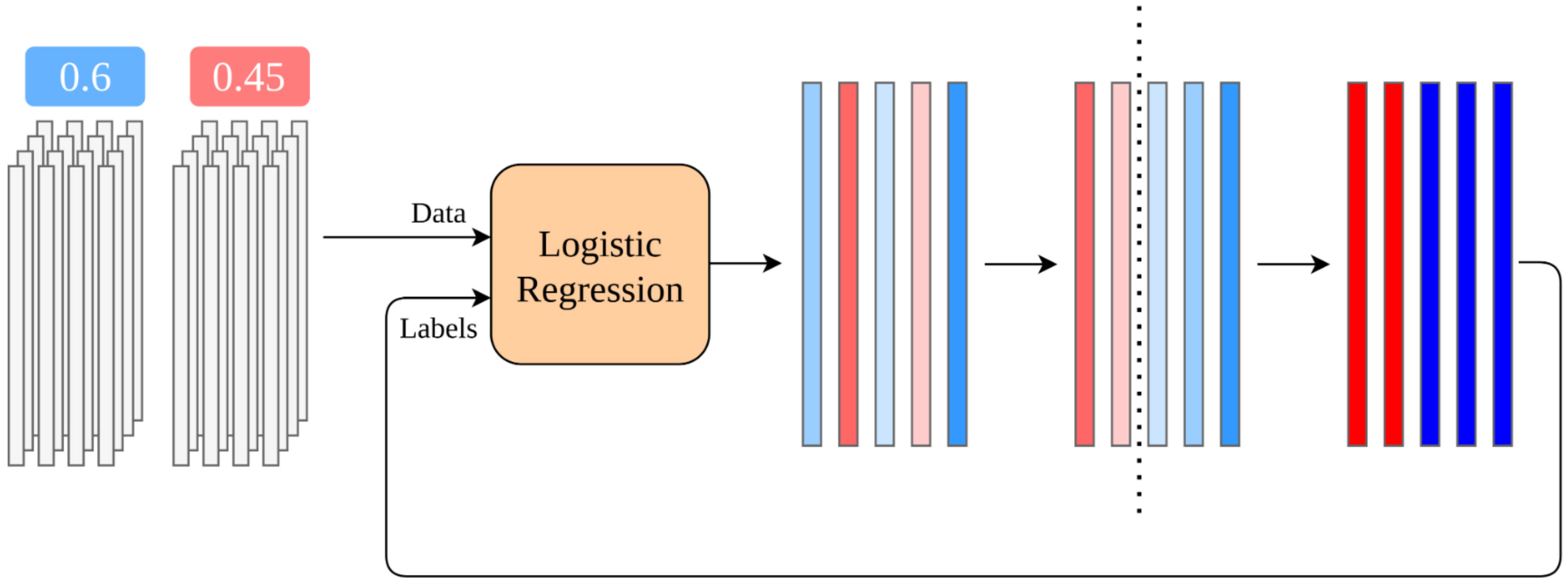
- The first implementation under MPC of an LLP model
- Mix of out-of-the-box components and custom implementations

Learning from Label Proportions (LLP)

- Each user uploads an *unlabeled* 1,034-element vector every day
 - Number of visits to the top 517 sites
 - Number of times referred to the top 517 sites
- Unlabeled vectors are grouped by state
- Each state has a ground-truth label
- Train on aggregate ground truth



The Plaintext Algorithm



Repeat Until Convergence

Translating to MPC

```
initialize_labels()

repeat until convergence:
    model = train(data, labels)
    predictions = inference(model, data)

    for each state:
        compute_threshold(truth[state], predictions[state])

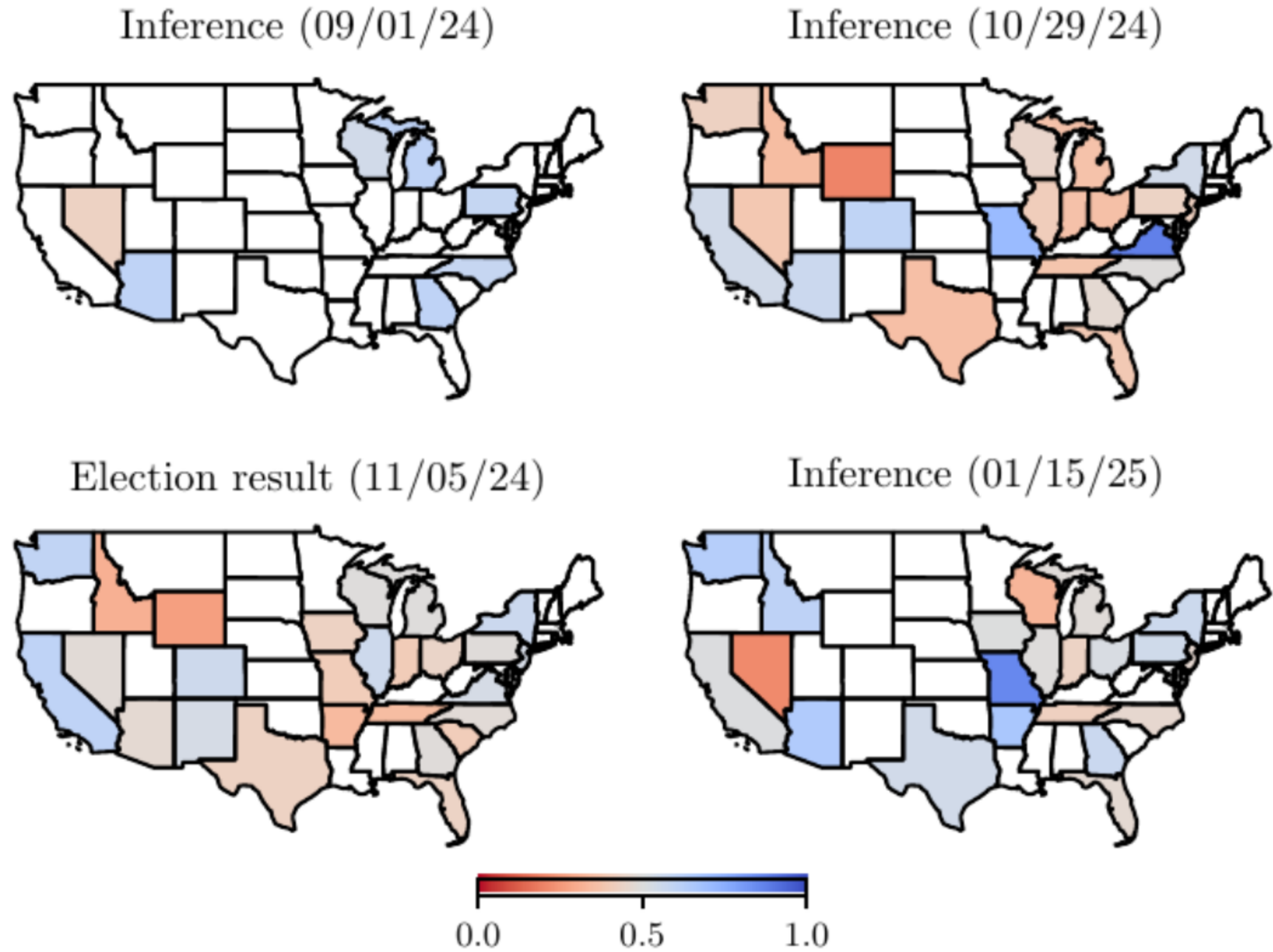
    for each user:
        labels[user] = predictions[user] >= thresholds[state]
```

- Plaintext initialization
- Training and inference as black boxes
- Computing thresholds requires oblivious sorting
- Updating labels requires oblivious comparison

Convergence Check

- Vector oblivious comparison
- Sum the comparison results
- Compare sum with convergence threshold

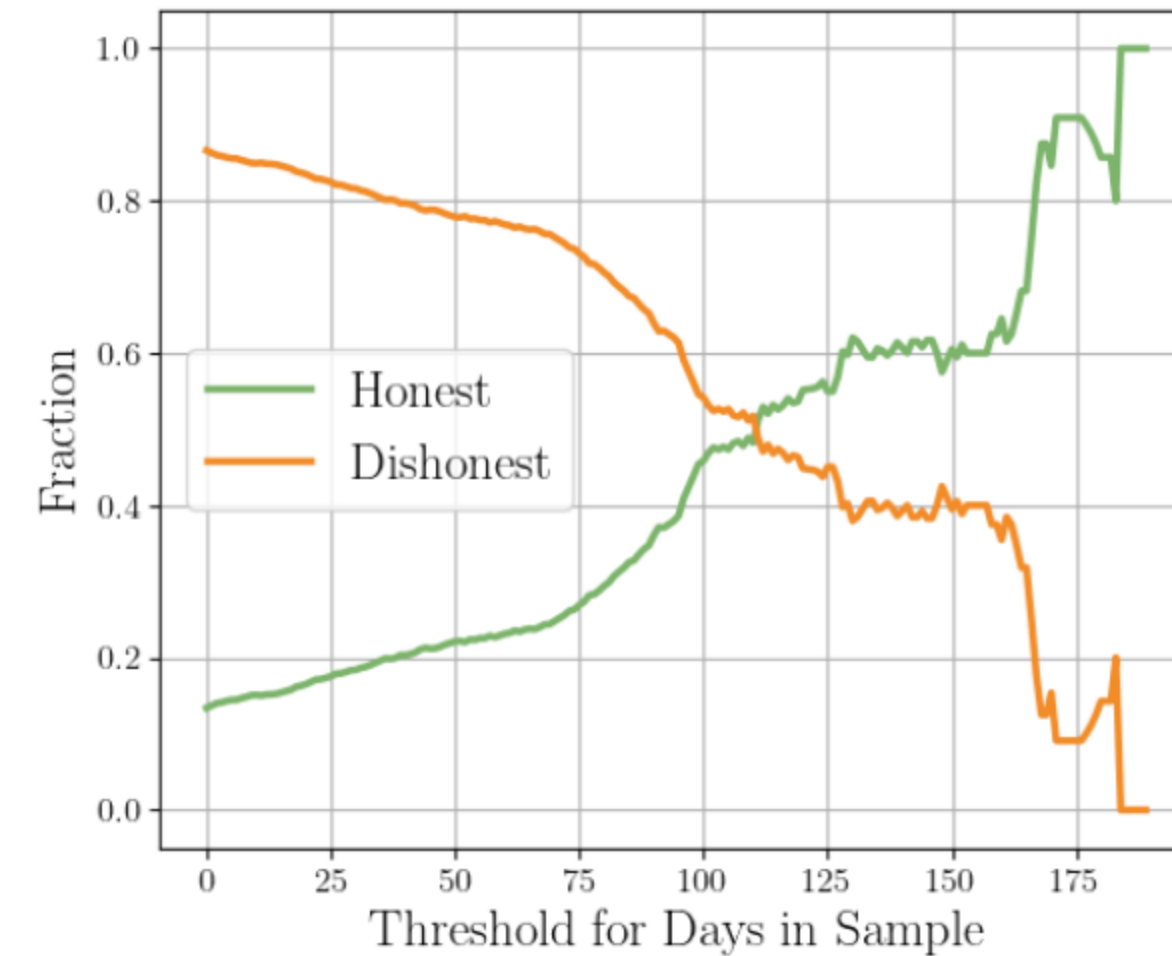
Preliminary Results



Lessons Learned and Future Directions

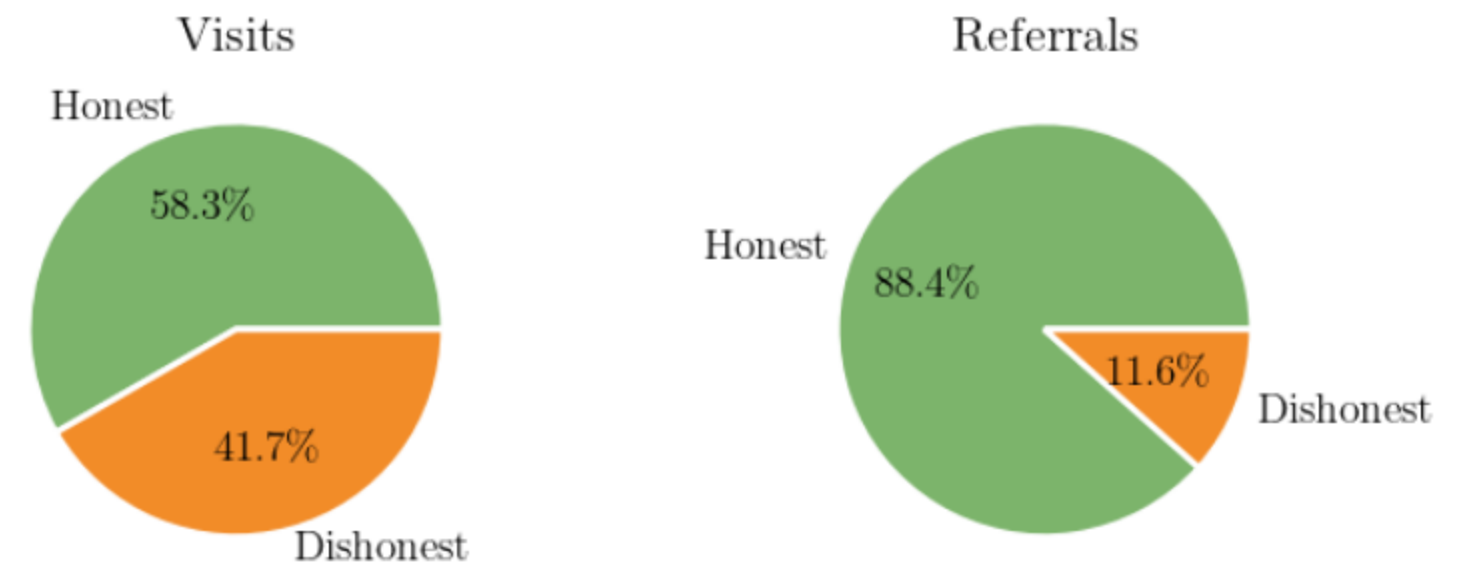
Data Integrity Matters

- Initial trouble with dishonest location reporting
- Validation with IP addresses and geolocation
- Results were mixed
 - Fully verified 15% of users
- Digging deeper on the data
 - Users in the sample for longer are more honest
 - Honest users contribute much richer data



Lesson: Validating and enforcing user honesty should be a priority in future deployments.

Lesson: Our data is surprisingly robust to dishonest users.



Strengthening the Threat Model

- AWS as a single point of failure
- Reduce or eliminate trust in the core computation
- Anonymous payments

Thank You!

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Payment

- Users are paid and recruited through Mechanical Turk
 - Requires knowledge of non-anonymous MTurk ID
 - Which users on which days
- BU IRB process took too long
 - BU was not allowed to hold personally identifiable information

The logo for Amazon Mechanical Turk, featuring the word "amazon" in white with its signature arrow, followed by "mechanical turk" in orange.

Communication by email

Data Partition



Input

Web Browsing Data

Registered IDs

Output

Aggregation by ID

ID Set per Day

Calculating Payments

WashU needed to know which users sent data on which days.

Users send private tags.

`tag = Hash(id || date)`

≈ 96 bits of entropy

- BU sends all collected tags to WashU
- WashU computes all combinations of users and dates and checks for matches

Computing Anonymous IDs

BU wants to connect data from the same user over multiple days.

WashU maps IDs to anonymized IDs between 1 and n .

- WashU sends all possible tags for all anonymous IDs
 - All (**date**, **hash**) combinations
- BU aggregates data from same anonymous ID for each week